

MUHAMMAD HASSAN

Artificial Intelligence · Machine Learning · Deep Learning | BS Computer Science Student



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PERSONAL STATEMENT

Final-year Computer Science student at National Skills University, Islamabad, targeting postgraduate scholarship opportunities in Artificial Intelligence and Machine Learning. Possesses strong academic grounding with a 3.41/4.00 CGPA alongside upcoming final-year research in multimodal deep learning, spatio-temporal forecasting, and computer vision. Demonstrates proven technical adaptability, leadership in large-scale academic initiatives, and professional web engineering background. Driven to advance research in scalable AI systems and spatial intelligence.

EDUCATION AND TRAINING

Bachelor's in Computer Science National Skills University, Islamabad

2023 – 2027 | CGPA 3.41 / 4.00

Focus: Artificial Intelligence, Machine Learning, Data Structures, Algorithms, Software Engineering.

Intermediate (ICS, Computer Science) Harvard College of Commerce & Science

2021 – 2023 | 81.27%

Matriculation (Computer Science) Shining Star Public School

2021 | 85%

RESEARCH & ACADEMIC PROJECTS

Multimodal Wildfire Risk Forecasting National Skills University, Islamabad

Upcoming Final Year Research Project | Scheduled Start: FYP 2026–2027

- ◆ Selected and designed as the upcoming undergraduate Final Year Project (FYP), investigating proactive wildfire risk forecasting by fusing multi-source remote sensing data with meteorological time-series records.
- ◆ Formulated a research methodology to build a multimodal fusion architecture combining Vision Transformers (ViT) for satellite environmental features (Sentinel-2/Landsat) and LSTM networks for historical weather patterns (NOAA) to project localized risk up to 5 days ahead.
- ◆ Planning integration of NASA FIRMS active fire datasets, spatial feature engineering via GeoPandas and Rasterio, and model explainability using SHAP.
- ◆ Targeting a production-ready system architecture with FastAPI, Docker, and an interactive Streamlit and Folium geospatial dashboard.
- ◆ Core Technologies: Python, PyTorch, Google Earth Engine, Sentinel-2/Landsat, NASA FIRMS, NOAA Data, GeoPandas, Rasterio, ViT (Hugging Face), LSTM, Scikit-learn, SHAP, Streamlit, Folium, FastAPI, Docker.

TECHNICAL & PERSONAL SKILLS

- **Programming Languages:** Python (NumPy, Pandas, PyTorch, TensorFlow), Java, C/C++, JavaScript, TypeScript
- **Core AI / ML Domains:** Multimodal Fusion, Computer Vision (Vision Transformers / ViT), Time-Series Forecasting (LSTM), Deep Learning, Geospatial AI, Explainable AI (SHAP)
- **Remote Sensing & Data Tools:** Google Earth Engine, Sentinel-2, Landsat, NASA FIRMS, NOAA Weather Data, Rasterio, GeoPandas, Folium
- **Frameworks & Deployment:** FastAPI, Docker, Streamlit, Hugging Face Transformers, Scikit-learn, React.js, Node.js, Express.js, MongoDB, Vite, Tailwind CSS
- **Developer Platforms:** Git, GitHub, Figma, Vercel, VS Code

- **Interpersonal & Leadership:** Technical Communication, Academic Research Writing, Event Logistics Management, Public Speaking

CERTIFICATIONS & CREDENTIALS

Huawei Certified Artificial Intelligence — Huawei (2026 – In Progress)

Specializing in cloud-native AI integration, scalable deep learning pipelines, and model deployment strategies.

Web Development Fundamentals — IBM SkillsBuild (2026)

Mastered web standards, modern frontend frameworks, and modular software design patterns.

VOLUNTEER & EXTRA-CURRICULAR LEADERSHIP

Logistics Lead — University Annual Party Organizing Committee (Mar 2026)

- ◆ Directed event operations and vendor coordination for 1,000+ attendees while leading a 25-member execution team.

Volunteer Coordinator — University Career Fair Organizing Team (Jul 2026)

- ◆ Supervised a 25-member volunteer team, managing schedules, task allocations, and operational logistics.

Timekeeper — National Debate Competition (Nov 2025)

- ◆ Managed official timing and speaker allocations across 171 participating universities nationwide.

Volunteer Invigilator — University Entry Test Support Team (Jul 2026)

- ◆ Assisted in test administration and exam integrity as part of a 50-member invigilation group.

CONFERENCES & ACADEMIC PARTICIPATION

- ◆ Participant, NUMLogic Competition, National University of Modern Languages (2025)
- ◆ Participant, SkillVerse Competition, National Skills University Islamabad (2025)
- ◆ Attendee, 3-day Workshop on Civic Responsibilities and Life Skills, NSU Islamabad (2025)
- ◆ Attendee, International Day of Clean Energy Event, NSU Islamabad (2025)

WORK EXPERIENCE

Freelance Web Developer — Fiverr (2025 – Present)

- ◆ Gathered technical requirements and architected custom React/JavaScript applications for global clients.
- ◆ Maintained a 100% completion rate and a 5.0 client feedback rating through reliable execution and clean code delivery.

REFERENCES

- **Dr. Shahid Anwar** — Assistant Professor, Department of Computing and Emerging Technologies, National Skills University Islamabad | shahidanwar@nsu.edu.pk
- **Dr. Huma Israr** — Assistant Professor, Department of Computing and Emerging Technologies, National Skills University Islamabad | hisrar.dphd17seecs@seecs.edu.pk